

Demographic Predictors of Electoral Volatility in U.S. Battleground Counties

A Machine Learning and Statistical Study of 2016–2024 U.S. Presidential Election Shifts

Sam Goodell, Antoni Czolowski, and Will Creager

University of Colorado Boulder, Boulder, CO, USA

{Sam.Goodell, Antoni.Czolowski, Will.Creager}@colorado.edu

ABSTRACT

We examine whether county-level demographics can be used to explain and predict electoral volatility across Pennsylvania, Michigan, and North Carolina, three electorally pivotal swing states in contemporary U.S. politics. Using American Community Survey variables and presidential election outcomes from 2016, 2020, and 2024, we constructed a county-level volatility measure based on changes in vote margin across election cycles. We then applied classification and regression methods, including Random Forest, XGBoost, support vector machines, and multiple linear regression. The results demonstrate that demographic variables provide meaningful predictive signals, particularly when volatility is framed as a binary classification problem rather than a five-class task. XGBoost performed best overall in multiclass classification, while Random Forest achieved the strongest binary results. Racial diversity and housing-related measures consistently emerged as important predictors, but cross-state validation produced near-random performance, indicating that the relationship between demographics and volatility is highly state-specific. As a practical extension, we developed a state-specific Campaign Priority Score that combines structural persuadability, predicted volatility, and electoral weight to rank counties within each battleground state. This framework illustrates how the paper’s predictive findings can be translated into an applied county-level targeting tool.

CCS CONCEPTS

Computing methodologies -> Machine learning; Computing methodologies -> Supervised learning by classification; Applied computing -> Social and behavioral sciences.

KEYWORDS

electoral volatility, swing states, county-level analysis, demographics, classification, regression, clustering

REFERENCE FORMAT

Sam Goodell, Antoni Czolowski, and Will Creager. 2026. Demographic Predictors of County-Level Electoral Volatility in Three U.S. Battleground States: Using Census Demographics to Model Volatility Across Recent Presidential Elections. Final project report for graduate course, University of Colorado Boulder. 10 pages.

1 INTRODUCTION

Amid negative campaigning, partisan media, and increasingly divided online discourse, the polarization of the U.S. political environment has been well documented in recent election cycles. Abramowitz (2022) finds that the correlation between party identification and ideological identification rose from 0.36 in 1984 to 0.68 in 2020 - a substantial rise in partisan-ideological consistency within the electorate. The current polarized environment raises a practical problem for campaign strategy: if much of the electorate is already politically entrenched, large portions of campaign resources can be misallocated to stubborn voters. Investments in persuasion are unlikely to pay off uniformly across counties, and untargeted spending may produce little electoral return. As a result, campaigns must focus on identifying specific places where volatility, competitiveness, and demographic change create genuine opportunity for successful intervention. The present study is motivated by this problem of strategic efficiency. Rather than treating battleground geography as uniformly competitive, we examine whether county-level demographics can help identify where electoral volatility is most likely to occur, and therefore where campaign resources have the highest marginal value.

Existing work on U.S. presidential elections has focused on forecasting aggregate outcomes rather than identifying the local dynamics that underlie strategic electoral opportunity. Abramowitz's (1988) "Time for a Change" model famously used presidential approval, economic growth, and duration of incumbent party control to predict national popular vote share, while Norpoth's Primary Model emphasized national primary performance (Norpoth, 2004). Although these models have proved useful in forecasting election outcomes, they rely on national indicators and provide limited insight into where campaign resources are likely to produce substantial marginal returns.

More recent work has moved toward finer geo-political analysis. Chen and Patel (2021), for example, narrowed their focus to swing states and used hierarchical clustering to identify distinct state archetypes; Crane, Koch, and Lin (2024) examined county-level voting behavior and found that stock market participation was associated with incumbent support under favorable market conditions. However, state-level approaches are still too broad to capture within-state volatility, and narrower county-level studies often isolate a single predictor at a time. As a result, it remains unclear whether a broader set of county-level demographic characteristics can systematically explain and predict electoral volatility in a way that is useful for campaign prioritization. This study addresses that gap by examining whether demographic patterns can identify counties where volatility, and therefore strategic opportunity, is most likely to emerge across three electorally pivotal swing states: Pennsylvania, Michigan, and North Carolina. To approach this question, we use a combination of classification and regression methods to analyze county-level electoral volatility across multiple levels of scope. In addition to comparing fine-grained and broader volatility formulations, we examine whether these demographic relationships hold in pooled, cross-state, and within-state settings.

Our results show that demographic variables do contain meaningful signals for electoral volatility, though signals are stronger for broad high-versus-low volatility classification than for fine-grained multiclass prediction. Across models, racial diversity and housing-related variables emerge as especially important predictors. At the same time, cross-state generalization is weak, suggesting that the relationship between demographics and volatility is highly state-specific. These findings suggest that demographic data can help campaigns identify broad windows of strategic opportunity, while also highlighting the limits of a single pooled model for national-style targeting.

2 METHODS

To test our hypothesis that voting volatility within counties can be predicted and explained using measurable demographic patterns, we create our own target metric, `vol_z_abs_sum`, which measures total swing within counties between the 2016/2020 and 2020/2024 elections. The study utilizes various classification techniques and multiple linear regression to explain and predict these scores.

2.1 Data Sources

The study is based on demographic data and our own metric, `vol_z_abs_sum`, which we calculated using MIT MEDSL Election Returns from Harvard Dataverse. Demographic data was pulled from two different APIs (IPUMS NHGIS and Census Bureau). Several overlapping variables between the data sources were compared, with no discrepancies found. The data was pulled for all 250 counties (100 North Carolina, 83 Michigan, and 67 Pennsylvania) across Pennsylvania,

Michigan, and North Carolina in the past three U.S. Presidential election years (2016, 2020, 2024). We then combined all three datasets, merging on `county_fips` and `election_year`.

2.2 Target Construction

2.2.1 *vol_z_abs_sum*

We created our target metric, `vol_z_abs_sum`, for each county by first finding the difference between their 2020 and 2016 Democratic vote margin, subtracting the mean difference among all counties, and dividing by the standard deviation to get their shift from those years as a z-score. We then repeated the process for their 2024 and 2020 Democratic vote margin and added both z-scores together.

2.2.2 5-class Volatility Bins

For our classification models, we binned the counties into 5-class quintiles based on their `vol_z_abs_sum` score. The resulting classes, each containing 50 counties, are Highly Stable (-1.675 median `vol_z_abs_sum`), Stable (-0.954), Moderate (-0.314), Volatile (0.607), and Highly Volatile (2.076).

2.2.3 Binary Target

We also created a binary class target for our classification models. The High-Volatility class contains the 100 counties in the Highly Volatile and Volatile bins, while the Low Volatility class contains the 150 counties from the remaining three bins.

2.3 Preprocessing and Feature Selection

2.3.1 Log Transformations and Scaling

We applied a $\log(1 + x)$ transformation to five heavily right-skewed predictors to bring their distributions closer to normal. After applying log transformations where appropriate, all continuous predictors were standardized to prepare the data for modeling.

Table 1. Skewness of predictors before and after log transformation.

Variable	Skewness Before	Skewness After	Reduction
population_density	11.09	0.36	97%
total_population	4.11	0.27	93%
median_home_value	1.49	0.39	74%
median_gross_rent	1.36	0.68	50%
median_household_income	1.05	0.28	73%

2.3.2 Final Feature Selection

We dropped the predictors `pct_renter_occupied`, `pct_no_hs_diploma`, `pct_white`, and `land_area_sqmi` to reduce multicollinearity. We selected 29 demographic predictors from six thematic groups to use for our models. Election variables were excluded to prevent leakage.

Table 2. Thematic feature groups used in the final model set.

Group	Features	n
Race/Ethnicity	<code>pct_black</code> , <code>pct_asian</code> , <code>pct_two_or_more_races</code> , <code>pct_hispanic</code> , <code>pct_non_hispanic_white</code> , <code>race_entropy_norm</code>	6
Urbanization	<code>log_total_population</code> , <code>log_population_density</code> , <code>pct_drive_alone</code> , <code>pct_carpool</code> , <code>pct_public_transit</code> , <code>pct_work_from_home</code>	6

Education	pct_hs_or_higher, pct_bachelors_plus	2
Housing	log_median_gross_rent, log_median_home_value, pct_owner_occupied	3
Economic	log_median_household_income, pct_below_poverty, pct_income_under_25k, pct_income_50k_100k, unemployment_rate	5
Household/Age	median_age, pct_senior_65plus, pct_young_adult_18_24, pct_foreign_born, pct_family_households, pct_married_couple, pct_living_alone	7

2.4 Modeling Framework

To evaluate whether county-level demographic variables could predict electoral volatility, we applied three classification models—Random Forest, XGBoost, and support vector machines (SVM)—alongside a multiple linear regression (MLR) model. For the classification models, hyperparameters were selected using inner 3-fold cross-validation. The final Random Forest model used `n_estimators = 300`, `max_depth = 8`, `min_samples_leaf = 5`, `max_features = sqrt`, and `class_weight = balanced`. The final XGBoost model used `n_estimators = 300`, `max_depth = 5`, `learning_rate = 0.1`, `subsample = 0.8`, `colsample_bytree = 0.8`, `reg_alpha = 0.1`, and `reg_lambda = 5`. The final SVM model used `C = 1`.

To complement the classification analysis, we also estimated a multiple linear regression model using the continuous volatility outcome. The MLR model used forward selection with mean absolute error as the selection criterion under 5-fold cross-validation to determine the final feature subset, after which the model was fit using ordinary least squares (OLS).

2.5 Evaluation Strategy

We evaluated our classification models using nested cross-validation, with an inner 3-fold loop for hyperparameter tuning and an outer 5-fold loop for performance estimation. Models were compared with F1-macro, accuracy, and ROC-AUC. We also conducted a leave-one-state-out technique (LOSO; training on two states, testing on the third) on the XGBoost model, using the same metrics for evaluation.

For our MLR model, we use an 80/20 split of the dataset to create training and test sets. We fit the model using OLS on the training set, then ran predictions on the test set. We evaluated the predictive power of the model using the RMSE and R^2 .

3 RESULTS AND DISCUSSION

The following analysis examines the extent to which county-level demographic characteristics explain and predict electoral volatility across Pennsylvania, Michigan, and North Carolina. We first evaluate predictive performance across the principal modeling approaches, then identify the demographic features most consistently associated with volatility, assess whether these relationships generalize across states, and finally present the applied ranking framework derived from these results.

3.1 Predictive Performance of Demographic Models

3.1.1 5-Class Results

The predictive results indicate that county-level demographic characteristics contain meaningful signals for electoral volatility, although the predictive power of the signals depends heavily on how the prediction task is defined. In the five-class setting, where counties were grouped into ordered volatility categories from Highly Stable to Highly Volatile, overall performance was modest across models. The best-performing classifier was XGBoost, which achieved 0.388 accuracy and a macro F1 score of 0.387, slightly outperforming Random Forest and linear SVM. Random Forest produced comparable overall accuracy at 0.368, but achieved the strongest ROC-AUC at 0.803; linear SVM performed similarly in macro F1 at 0.364. These results suggest that demographic data can distinguish broad patterns of volatility well above the random-guess baseline, but fine-grained separation among the middle volatility classes remains difficult.

Table 3. Performance metric comparison across multiclass classification models.

Model	Accuracy	F1-macro	ROC-AUC
XGBoost	0.388	0.387	0.783
SVM (Linear)	0.368	0.364	0.765
Random Forest	0.368	0.362	0.803

The primary limitation of the five-class models was the density of the middle classes, where counties exhibited overlapping demographic profiles and relatively small differences in underlying volatility. For example, Random Forest achieved class-specific F1 scores of 0.46 for Highly Stable and 0.59 for Highly Volatile, while XGBoost achieved 0.37 and 0.56 for those same classes. By contrast, the middle categories showed substantially weaker, more diffuse classification performance. The pattern implies that county-level demographics are more effective at identifying clearly stable and clearly unstable political environments than at making precise distinctions among intermediate levels of electoral volatility.

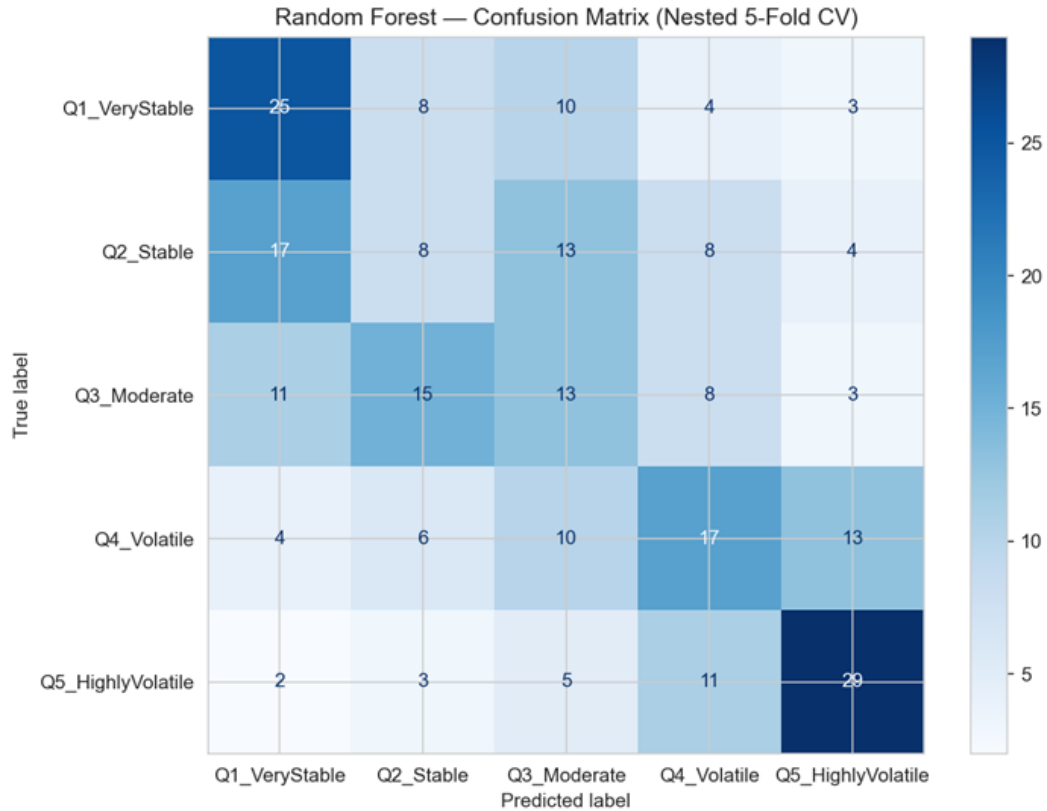


Figure 1. Random Forest confusion matrix shows strong diagonal performance at the most stable and most volatile class extremes, with greater confusion among middle classes.

Regression results reinforce this interpretation, though they indicate more limited predictive strength when volatility is modeled as a continuous outcome. The pooled multiple linear regression achieved an R^2 of 0.236 and an RMSE of 1.122 on the held-out test set, indicating that demographic variables explained some meaningful share of variance but left most volatility unexplained. Relative to the classification models, regression was more useful as an interpretive benchmark than as the strongest predictive tool. Its modest performance supports the broader conclusion that demographics capture part of the volatility process, but not enough to support precise continuous prediction on their own.

3.1.2 Binary Results

A five-class setting imposes a level of separation that may exceed the practical needs of campaign targeting, as the middle volatility classes are comparatively ambiguous and less distinct. In practice, campaign strategists are less interested in whether a county is moderately volatile; instead, they are attempting to identify where meaningful electoral movement is most probable and resource investment is least likely to be wasted. This suggests that a broader class framework may be more appropriate, particularly if demographic variables are strongest at distinguishing the tails of the volatility distribution. We therefore reframe the task as a binary classification problem, separating higher-volatility counties from lower-volatility counties to test whether predictive performance improves when the model is asked to identify only the more strategically relevant tail.

Model performance improved substantially under binary classification. Random Forest produced the strongest overall binary results, achieving 0.780 accuracy, 0.718 F1, and 0.828 AUC, outperforming XGBoost, RBF SVM, and linear SVM. XGBoost and RBF SVM remained competitive, but neither matched the balance of precision and recall achieved by Random Forest. At the same time, this improvement should be interpreted with appropriate caution: the binary task carries a much higher baseline than the five-class setting, where random guessing would produce only 20% accuracy. The binary majority-class baseline is 60%, so an increase in headline performance is expected even without a major gain in substantive signal. Nevertheless, relative to the five-class setting, this improvement suggests that county-level demographic characteristics are more effective at distinguishing broad volatility risk than at assigning counties to narrowly separated categories. The binary formulation appears to better capture the practical structure of the problem.

Table 4. Binary classification performance summary by model.

Model	Accuracy	F1	Precision	Recall	AUC
Random Forest	0.780	0.718	0.726	0.690	0.828
XGBoost	0.748	0.667	0.718	0.610	0.803
SVM (RBF)	0.760	0.674	0.724	0.630	0.789
Logistic Regression	0.680	0.604	0.598	0.610	0.745
SVM (Linear)	0.688	0.610	0.610	0.610	0.734

3.2 Key Predictors of Electoral Volatility

Across the classification and regression analyses, a fairly consistent set of demographic predictors emerged as most closely associated with electoral volatility. The most prominent included racial diversity and housing-related measures, with state-specific roles played by poverty, education, and population density. While the relative importance of individual features varied by model, the broader pattern was stable: volatility was strongly associated with counties experiencing demographic complexity, economic transition, or both.

Racial diversity, measured through `race_entropy_norm`, was among the most consistent predictors of electoral volatility across the study. In the Random Forest analysis, permutation importance identified it as the most important feature. XGBoost results suggest racial diversity again ranked among the top predictors; dependence plots suggested a generally positive relationship between diversity and the probability of belonging to the Highly Volatile class. These results imply that counties with more heterogeneous racial composition were systematically more likely to exhibit unstable or shifting electoral behavior.

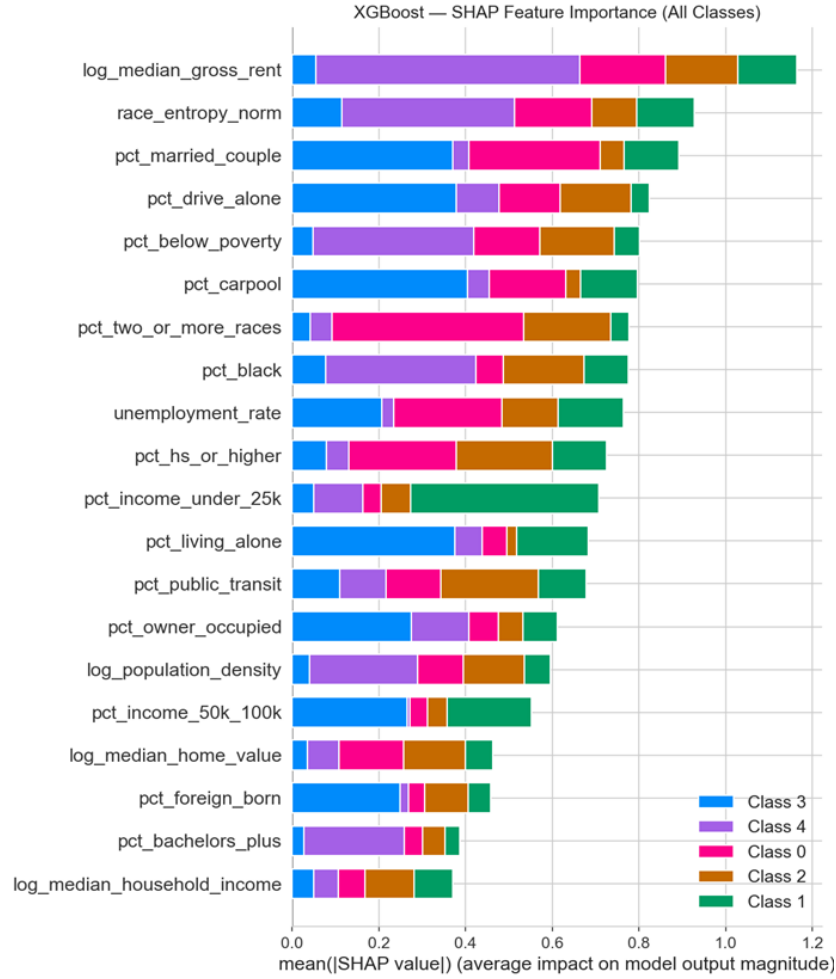


Figure 2. XGBoost SHAP feature importance across classes, highlighting housing cost and racial entropy among the most influential predictors.

Housing-related variables were similarly prominent, especially in the boosted models and regression analyses. `log_median_gross_rent` emerged as the dominant predictor in the XGBoost SHAP analysis with particular influence on the Highly Volatile class. Regression results reinforced the importance of housing measures, with both `log_median_gross_rent` and `log_median_home_value` among the retained predictors in the pooled model. These variables likely capture local economic transition, growth pressure, and changing residential patterns, all of which may coincide with electoral instability.

3.3 State-Level Generalizability

We sought to examine whether the relationship between county-level demographics and electoral volatility generalizes across battleground states or remains largely state-specific. The results indicate that cross-state generalizability was weak. The best-performing five-class model, XGBoost, fell to near-random levels in all three rotations in a leave-one-state-out (LOSO) setting. Accuracy ranged from 0.205 to 0.210 and macro F1 from 0.182 to 0.213, only marginally above the 0.200 random-guess baseline for a five-class task. These results suggest that demographic patterns associated with volatility do not transfer cleanly across state boundaries.

Within-state modeling produced stronger classification results. When binary XGBoost classifiers were estimated independently for Pennsylvania, Michigan, and North Carolina, performance improved substantially - the state-specific models produced AUC values of 0.838, 0.778, and 0.829 and F1 scores of 0.591, 0.636, and 0.736, respectively. The contrast between cross- and within-state performance indicates that demographics can meaningfully predict volatility, but primarily when the model is calibrated to the local structure of a given state rather than applied across states as a unified rule.

Table 5. Within-state classification performance summary.

State	n	LOSO F1	Within-State F1
Pennsylvania	67	0.209	0.591
Michigan	83	0.205	0.636
North Carolina	100	0.210	0.736

Feature importance patterns diverged in similar fashion. Pennsylvania's within-state volatility was driven most strongly by racial diversity, Michigan's by educational attainment, and North Carolina's by poverty, while housing cost was the only predictor that appeared consistently across all three states. The findings indicate that electoral volatility is shaped by state-specific demographic and political mechanisms rather than a uniform national pattern.

4 CONCLUSIONS AND FUTURE WORK

4.1 Limitations and Ethical Considerations

Several limitations should be noted. Political conditions are not static, and the demographic relationships captured here may shift across election cycles, requiring frequent model recalibration. Second, the analysis does not directly account for turnout, partisan enthusiasm, or mobilization effects, which may follow different dynamics from persuasion-oriented volatility. More broadly, the models omit campaign-specific influences such as candidate quality, local organizing strength, media environment, and issue salience. There are also ethical considerations associated with any applied targeting framework. A system designed to identify high-opportunity counties may encourage intrusive political marketing saturation, voter fatigue, and uneven exposure to political messaging across communities.

4.2 Discussion and Strategic Implications for Campaign Targeting

This study examined whether county-level demographic characteristics can explain and predict electoral volatility across Pennsylvania, Michigan, and North Carolina. The results indicate that demographic variables do contain meaningful signals, but that signal is neither uniform nor equally useful across all modeling tasks. In the five-class setting, predictive performance was modest, with the strongest results concentrated at the highly stable and highly volatile ends of the distribution. This pattern is substantively important because the Highly Volatile class represents the primary class of interest from a campaign perspective: counties in this group are the most likely to contain meaningful electoral movement and therefore the greatest potential return on persuasion-oriented resource allocation. Random Forest achieved a class-specific F1 of 0.59 for the Highly Volatile class while XGBoost achieved 0.56, indicating that the models retained practical value in identifying the counties of greatest strategic interest. To a lesser extent, performance on the Highly Stable class was also comparatively strong, which is useful in the opposite direction by detecting counties where campaign resources are less likely to produce meaningful returns.

When volatility was reframed as a broader binary distinction between high- and low-volatility counties, performance improved substantially, indicating that demographic characteristics are more effective at supporting broader distinctions than finely partitioned class assignments. Across models, racial diversity and housing-related measures emerged as the most consistent predictors of volatility.

Results suggest the relationship between demographics and electoral volatility is highly state-specific. Cross-state analyses produced near-random results, while within-state models performed much better, indicating that the demographic mechanisms associated with volatility do not generalize cleanly across battleground states. Although some variables, particularly housing cost, appeared across multiple settings, the dominant predictors differed meaningfully across Pennsylvania, Michigan, and North Carolina. These findings suggest that county-level demographic patterns are informative, but only when interpreted within distinct state contexts rather than as part of a single national relationship.

Our findings contribute to topical literature by shifting attention from aggregate electoral forecasting toward the local demographic conditions associated with electoral instability. More broadly, they suggest that electoral volatility is better understood as a state-contingent process shaped by different combinations of demographic diversification, housing-market change, and local socioeconomic structure rather than a uniform battleground phenomenon. In this respect, the

paper's main contribution is to show both the value and the limits of demographic modeling: models can identify meaningful patterns in county-level volatility, but their usefulness depends on the level of classification, the geographic scope of application, and the degree to which local political context is taken into account.

Future research could extend this analysis. First, subsequent work could move more explicitly from explanation to prioritization by developing and validating county-ranking frameworks designed for campaign allocation. The predictive feature set could be broadened beyond demographics to include variables such as employment composition, industry structure, local political leadership, and additional economic indicators that may better capture the mechanisms underlying electoral instability. Finally, future studies could integrate turnout dynamics alongside margin-based volatility, allowing researchers to distinguish between counties that are electorally unstable because voters change preferences and those that are unstable because voting rates fluctuate across election cycles.

5 APPENDIX

5.1 Practical Extension: Foundations for Applied Ranking Framework

To translate the modeling results into a practical targeting tool, we constructed a Campaign Priority Score (CPS) to identify counties that appear most persuadable within each battleground state. The score is intended to identify counties that appear both politically movable and electorally meaningful within each battleground state.

The score is built from three components: persuadability, electoral weight, and predicted volatility. Formally defined as:

$$CPS_i = 0.25S_{pers,i} + 0.25S_{wt,i} + 0.50S_{vol,i}$$

$S_{wt,i}$ denotes the normalized, within-state electoral-weight component; $S_{vol,i} = p_{high-vol,i}$ denotes the predicted probability from a state-specific Random Forest model that county i belongs to the high-volatility class. The electoral-weight term is based on each county's share of the statewide vote, transformed by a square root and then scaled within state. The volatility term is generated using out-of-fold predictions, so that each county is scored by a model that did not train on that county itself. Consistent with the paper's broader findings, the volatility probabilities are derived from within-state models rather than a single pooled classifier.

The persuadability component is defined as a transparent linear structural index informed by the earlier feature-importance analysis:

$$S_{pers,i} = 0.40D_i + 0.25P_i + 0.20(1 - I_i) + 0.15N_i$$

D_i is the normalized racial diversity measure, P_i is the normalized poverty rate, I_i is normalized log median household income, and N_i is normalized log population density. Under this calculation, higher diversity, higher poverty, lower income, and greater population density all increase the persuadability score. These features were selected due to their relevance in the earlier XGBoost feature-importance (SHAP). The weighting scheme reflects a first-pass attempt to summarize those signals while keeping the applied measure interpretable.

This ranking framework is intended to operationalize the main findings of the paper rather than replace them. The weighting scheme gives the greatest emphasis to model-based volatility, reflecting the paper's broader conclusion that demographics are most useful for identifying counties that are broadly more likely to exhibit electoral instability. Persuadability and electoral weight each receive smaller but still meaningful roles, ensuring that the score reflects both the structural profile of a county and potential electoral impact. Counties are ranked within each state rather than across all three states together, consistent with the earlier finding that volatility mechanisms do not generalize cleanly across state lines.

This score should be interpreted as an initial applied framework rather than a finalized targeting model. The current persuadability term is intentionally linear for transparency, even though the earlier SHAP and dependence analyses suggest that some of these relationships may be threshold-based, stepwise, or otherwise nonlinear. Future work should therefore treat this score as a "first pass" and explore alternative functional combinations, additional predictors, and further validation to better capture the underlying dynamics of persuadability across counties.

Table 6. Top 5 Campaign Priority Scores (PA)

State	County	State Rank	Campaign Priority Score
Pennsylvania	Philadelphia	1	0.838
Pennsylvania	Allegheny	2	0.800
Pennsylvania	Delaware	3	0.736
Pennsylvania	Montgomery	4	0.690
Pennsylvania	Lehigh	5	0.633

Table 7. Top 5 Campaign Priority Scores (NC)

State	County	State Rank	Campaign Priority Score
North Carolina	Halifax	1	0.697
North Carolina	Wake	2	0.689
North Carolina	Lenoir	3	0.688
North Carolina	Washington	4	0.672
North Carolina	Richmond	5	0.669

Table 8. Top 5 Campaign Priority Scores (MI)

State	County	State Rank	Campaign Priority Score
Michigan	Oakland	1	0.797
Michigan	Wayne	2	0.793
Michigan	Washtenaw	3	0.720
Michigan	Kent	4	0.712
Michigan	Ingham	5	0.654

REFERENCES

- [1] Alan I. Abramowitz. 1988. An improved model for predicting presidential election outcomes. *PS: Political Science & Politics* 21, 4 (1988), 843-847.
- [2] Alan I. Abramowitz. 2022. The polarized American electorate: The rise of partisan-ideological consistency and its consequences. *Political Science Quarterly* 137, 4 (2022), 645-676. <https://doi.org/10.1002/polq.13388>
- [3] Mason Chen and Saloni Patel. 2021. A statistical study of swing state behavior in the 2020 United States presidential election. *American Statistical Association*, 9 pages.
- [4] Alan D. Crane, Andrew Koch, and Leming Lin. 2024. Real effects of markets on politics: Evidence from U.S. presidential elections. *American Economic Review: Insights* 6, 1 (March 2024), 73-88. <https://doi.org/10.1257/aeri.20220240>
- [5] Helmut Norpoth. 2004. From primary to general election: A forecast of the presidential vote. *PS: Political Science & Politics* 37, 4 (October 2004), 737-740. <https://doi.org/10.1017/S1049096504045032>
- [6] U.S. Census Bureau, “American Community Survey 5-Year Estimates: Comparison Profiles 5-Year,” 2016/2020/2024, <<http://api.census.gov/data/{year}/acs/acs5>>, accessed on February 22, 2026
- [7] Steven Manson, Jonathan Schroeder, David Van Riper, Katherine Knowles, Tracy Kugler, Finn Roberts, and Steven Ruggles. *IPUMS National Historical Geographic Information System: Version 19.0* [dataset]. Minneapolis, MN: IPUMS. 2024. <http://doi.org/10.18128/D050.V19.0>
- [8] MIT Election Data and Science Lab. 2018. County Presidential Election Returns 2000-2024. <https://dataverse.harvard.edu/citation?persistentId=doi:10.7910/DVN/VOQCHQ>.